JULIAN ESTEBAN RINCÓN RODRÍGUEZ

Computer Science and Artificial Intelligence | Data Science · Machine Learning · Data Engineering
Bogotá, Colombia | julianer2002@gmail.com | +57 311 828 3342
julian-rincon.github.io | github.com/Julian-Rincon | linkedin.com/in/julian-esteban-rincón-rodriguez-1a05501b7

PROFESSIONAL SUMMARY

Final-year (9th of 9 semesters) Computer Science and Artificial Intelligence student focused on data science and machine learning. I clean and rebuild datasets, train models evaluated once on held-out data with confidence intervals, and build the SQL and PySpark pipelines that feed them on AWS. Experience delivering software to clients and working in teams of three to five. Seeking a six-month data science or data analytics internship for January – June 2027.

EDUCATION

Universidad Sergio Arboleda — Bogotá, Colombia | 2022 – Present
B.Sc. Computer Science and Artificial Intelligence | 9th of 9 semesters | GPA 4.04/5.0, with the last three semesters above 4.5/5.0.

Relevant coursework: Big Data and Data Engineering, Machine Learning, Neural Networks, Natural Language Processing, Modeling and Simulation.

SELECTED PROJECTS

SAVI — Real Estate Valuation Agent (Ames, Iowa) | 2026 | v1–v2 team of 4, v3 individual
Python · XGBoost · LightGBM · PyTorch (DQN) · AWS Lambda · EC2 · SageMaker · RAG

- Audited the team dataset, found that 93% of it was assessor records padded with constant columns, and rebuilt it from public sources: the De Cock sales (2,930) joined to 18,078 parcels, with FHFA and Zillow price indices.
- Reached out-of-fold R^2(log) = 0.921 and 7.8% MAPE with XGBoost (LightGBM: 0.912, 8.3%), and trained RL policies that approve, review, or reject valuations; on a held-out test of 378 sales none beat Value Iteration significantly (95% bootstrap interval), and I report it that way.
- Shipped it as an event-driven AWS pipeline (S3 → Lambda → EC2 → SageMaker Spot) at under US$0.05 per run, plus a web agent whose answers are 10/10 grounded in tool outputs on a 10-question evaluation.

Sentiment Analysis on Sentiment140 with MLflow | Sep. 2026 | Team of 5
Python · scikit-learn · TF-IDF · MLflow Model Registry · FastAPI · AWS EC2 · SageMaker

- Compared preprocessing, representation, and classifier stages under cross-validation against a dummy baseline, with every run tracked in MLflow; picked TF-IDF (uni+bigrams) with logistic regression after an ablation study.
- Retrained the champion on 1.36M tweets, evaluated it once on test (macro-F1 0.824), registered it in the MLflow Model Registry, and served it through a FastAPI endpoint on EC2 with error analysis.

ShopStream — AWS Big Data Pipeline | May – Jul. 2026
PySpark · SQL · AWS Lambda · EMR · Glue · RDS PostgreSQL · Flask

- Built an end-to-end pipeline for 2.5M e-commerce events: S3 ingestion with Lambda validation and quarantine, EMR/PySpark analytics for six KPIs with anomaly detection, Glue ETL, and a PostgreSQL star schema; 86% test coverage and CI/CD.

AWS Big Data Platform (Chinook / NYC Taxi) | Mar. – May 2026 | Team of 3
PySpark · SQL · Athena · EMR · S3 · Glue · Terraform · FastAPI

- Processed 94M NYC Yellow Taxi trips (12 months) on EMR with PySpark, fixing schema drift across Parquet files, and exposed them as a partitioned Athena table queried with SQL.

Ayuda Terremoto Colombia — Humanitarian Coordination | Aug. 2026 – Present
Python · FastAPI · PostgreSQL · React · IndexedDB

- Built an offline-first platform after the magnitude 7.4 Chocó earthquake, with fuzzy-match report deduplication, automatic activation from USGS data, and HXL export; 80+ pytest tests.

Additional projects: Phantom, a multi-agent coding workspace that routes tasks by measured model success (Rust, TypeScript); NEXUS, a Python AI assistant with 1,400+ tests; Dino Island, a generative NLP pipeline (team of 5); an NLP API with spaCy and FastAPI (93/100 in a live evaluation).

PROFESSIONAL EXPERIENCE

Full-Stack and AI Developer (Freelance) — Zafra Asesores Tributarios | Girardot, Colombia | Jun. 2026

- Built a sales CRM for a tax advisory firm, modeling clients, prospects, campaigns, and interactions in a FastAPI/SQLAlchemy backend with a React/Vite view of the prospect pipeline.
- Added an LLM layer (Groq, Llama 3.3 70B) that summarizes interactions, drafts emails, and scores leads, plus a lead-qualification bot on the WhatsApp Cloud API; handed over with technical documentation.

Full-Stack Developer — ReushiGo | Bogotá, Colombia | May 2026

- Built an ordering platform for a family coffee distributor (React/Vite admin panel, Express/Prisma backend, Telegram sales bot over a live 19-product catalog) with CI on GitHub Actions.

TECHNICAL SKILLS

Languages: Python (intermediate-advanced), SQL, C, Bash; TypeScript and JavaScript (basic)

Data analysis: Pandas, NumPy, Matplotlib, Power BI, Jupyter, SQL (PostgreSQL, Athena), bootstrap confidence intervals, cross-validation

Machine learning: scikit-learn, XGBoost, LightGBM, PyTorch, spaCy, MLflow

Data engineering and cloud: PySpark, AWS (S3, EMR, Glue, Athena, RDS, Lambda, EC2, SageMaker), GCP, Terraform, Docker (basic)

Tools: Git, GitHub Actions, pytest, FastAPI, Linux (Fedora as daily OS)

ACHIEVEMENTS

1st Place — Wisibilízalas International Competition, Universitat Pompeu Fabra | 2019

- Technical product manager and frontend/UX lead of the winning web application, representing my school from Ecuador in the Senior category at 17, against teams from Latin America and Europe.

1st Place — USABOT Robotics Challenge | November 2025

- Led the CAD design and sensor architecture of a smart irrigation system and delivered the pitch, winning against university teams.

CERTIFICATIONS

AWS Academy Graduate — Data Engineering | Amazon Web Services Academy | May 2026 (40 hours)

Database Programming with SQL; Database Design | Oracle Academy | 2024

LANGUAGES

Spanish: native. English: certified B2 overall (iTEP Academic-Plus, November 2025; Listening C1).